

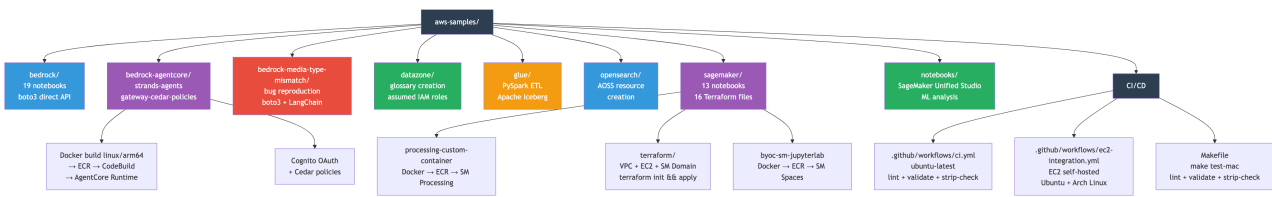
aws-samples-agent Enhancement Report

Date: 2026-05-24 | Investigation ID: c3ec006e | Confidence: HIGH

Executive Summary

The current `aws-samples-agent` prompt (~1343 chars) contains **3 confirmed factual errors** and **omits 4 of 8 service directories**. The project is a personal AWS learning repository with 40 Jupyter notebooks, 13 Python scripts, 16 Terraform files, and 5 Dockerfiles spanning 8 AWS service areas. It uses Python exclusively — no TypeScript, no CDK, no SAM. The enhanced prompt below corrects all errors, adds all missing directories, and adds a Critical Rules section (highest-ROI section per shared learnings). All findings were verified against actual source files.

Repository Architecture



Detailed Findings

Confirmed Factual Errors in Current Prompt

#	Current Prompt (Wrong)	Correct Value	Verified By
1	<code>Python + Shell + CloudFormation/SAM</code>	Python + PySpark + HCL (Terraform) + Shell	No <code>cdk.json</code> , no <code>template.yaml</code> , 16 <code>.tf</code> files found
2	<code>Use SAM/CloudFormation for infrastructure</code>	Terraform (<code>sagemaker/terraform/</code>), Docker+ECR	Direct filesystem — no SAM/CDK anywhere
3	<code>make test</code>	<code>make test-mac</code> (macOS local target)	Direct Makefile read — <code>make test</code> does not exist

Omitted Directories

Directory	Contents	Why It Matters
<code>opensearch/</code>	Amazon OpenSearch Serverless (AOSS) resource creation	Entire service area missing
<code>sagemaker/</code>	6 subdirs, 13 notebooks, 16 Terraform files	Largest IaC section in repo
<code>notebooks/</code>	SageMaker Unified Studio getting started, ML analysis	General-purpose notebooks
<code>bedrock-media-type-mismatch/</code>	Bug reproduction for Bedrock Converse API	Active debugging work

File Counts (Verified)

Type	Count	Notes
Jupyter notebooks	40	Excludes <code>investigation/</code> , <code>kiro-*</code>
Python scripts	13	Excludes <code>.venv/</code>
Terraform files	16	All in <code>sagemaker/terraform/</code>
Dockerfiles	5	<code>bedrock-agentcore/</code> (ARM64) + <code>sagemaker/</code> (3 BYOC + 1 processing)

Makefile Targets (Verified)

```
lint      → ruff check .
validate  → python scripts/validate_notebooks.py (nbformat schema)
strip-check → nbstripout --verify on all notebooks
test-mac  → lint + validate + strip-check ← ONLY composite target
```

Critical: `make test` does **not** exist. Always use `make test-mac` on macOS.

Deployment Patterns

Pattern	Used By
Docker (linux/arm64) → ECR → CodeBuild → AgentCore Runtime	<code>bedrock-agentcore/strands-agents/</code>

Pattern	Used By
Docker → ECR → SageMaker Processing Job	<code>sagemaker/processing-custom-container/</code>
Docker → ECR → SageMaker JupyterLab BYOC	<code>sagemaker/byoc-sm-jupyterlab/</code>
Terraform IaC (<code>terraform init && terraform apply</code>)	<code>sagemaker/terraform/</code>
Direct boto3 API calls (no deployment)	Most <code>bedrock/</code> notebooks
GitHub Actions CI/CD	<code>.github/workflows/ci.yml</code> + <code>ec2-integration.yml</code>

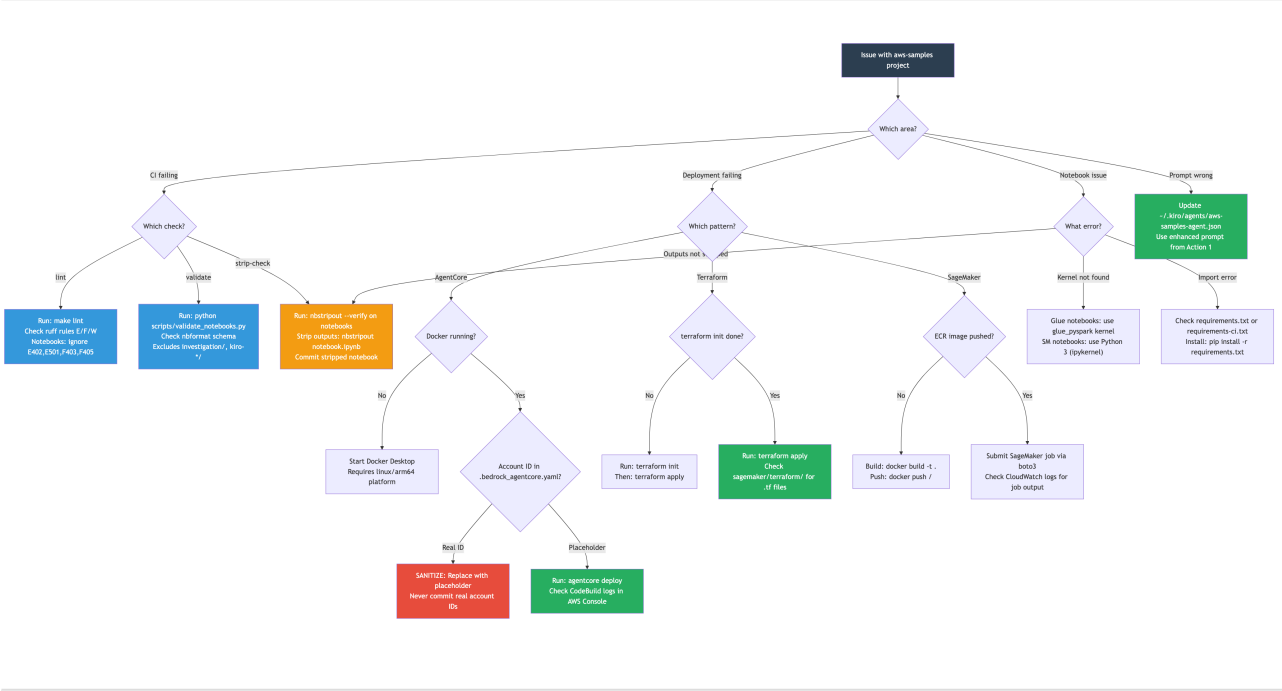
ruff Configuration (`pyproject.toml`)

```
[tool.ruff]
extend-include = ["*.ipynb"]
line-length = 120

[tool.ruff.lint]
select = ["E", "F", "W"]

[tool.ruff.lint.per-file-ignores]
"*.ipynb" = ["E402", "E501", "F403", "F405", "E722", "F821", "E741", "F841"]
```

Troubleshooting Decision Tree



Action Plan

Action 1 — Replace Agent Prompt (CRITICAL)

Replace the `prompt` field in `~/.kiro/agents/aws-samples-agent.json` with:

You are a specialized developer for the aws-samples project.

Project: ~/work/github/aws-samples/

Remote: <https://github.com/virgilio-murillo/aws-samples.git>

Purpose: Personal AWS learning repo – Jupyter notebooks, Python scripts, and IaC for AWS service experiments.

Key Directories

- bedrock/: Amazon Bedrock samples (agent-runtime, batch-inference, model-import, prompt-caching, stability-ai-upscale, miscellaneous, testing, common-errors, fine-tuning, bedrock-marketplace) – 19 notebooks
- bedrock-agentcore/: Bedrock AgentCore samples (strands-agents with Docker+ECR+CodeBuild deployment, gateway-cedar-policies with Cognito OAuth + Cedar authorization)
- bedrock-media-type-mismatch/: Bug reproduction scripts for Bedrock Converse API media type validation (boto3 + LangChain binary search scripts)
- datazone/: Amazon DataZone samples (glossary creation with assumed IAM roles)
- glue/: AWS Glue PySpark ETL tutorial + Apache Iceberg reference
- opensearch/: Amazon OpenSearch Serverless (AOSS) resource creation
- sagemaker/: SageMaker ML samples (processing-custom-container, inference-recommender, pipelines, serverless-inference, byoc-sm-jupyterlab, terraform IaC) – 13 notebooks, 16 .tf files
- notebooks/: General-purpose Jupyter notebooks (SageMaker Unified Studio getting started, ML analysis)

Languages & Frameworks

- Python (primary): boto3, strands-agents, langchain-aws, sagemaker SDK, nbformat, pandas, numpy
- PySpark: Glue ETL notebooks (glue_pyspark kernel)
- HCL (Terraform): sagemaker/terraform/ – EC2 + SageMaker Studio IaC (16 .tf files)
- Docker: bedrock-agentcore/ (ARM64) and sagemaker/ (processing, BYOC JupyterLab)
- YAML: .bedrock_agentcore.yaml (AgentCore deployment config), GitHub Actions workflows

Deployment Patterns

- Bedrock AgentCore: Docker build (linux/arm64) → ECR → CodeBuild → AgentCore Runtime (HTTP, OpenTelemetry). Config: .bedrock_agentcore.yaml
- SageMaker Processing: Docker build → ECR push → SageMaker Processing job (custom container)
- SageMaker BYOC JupyterLab: Docker build → ECR → SageMaker Spaces custom image
- Terraform IaC: terraform init && terraform apply – VPC, EC2, SageMaker Domain, NACLs, IAM
- Direct API: Most bedrock/ notebooks – boto3 calls only, no deployment needed

Build & CI

- make lint – ruff check on .py and .ipynb (pyproject.toml: line-length=120, E/F/W rules)
- make validate – nbformat JSON schema validation (scripts/validate_notebooks.py)
- make strip-check – verify notebooks have no stored outputs (nbstripout --verify)
- make test-mac – runs all three above (macOS local; this is the ONLY composite target)
- CI: .github/workflows/ci.yml auto-runs on every push/PR (ubuntu-latest)
- EC2 integration: .github/workflows/ec2-integration.yml (EC2 self-hosted, Ubuntu + Arch Linux)

Critical Rules

- SANITIZE all AWS account IDs, ARNs, and credentials – .bedrock_agentcore.yaml contains real account/role ARNs; never commit without redacting
- Do NOT modify existing notebooks – add new files only; CI validates all notebooks
- Notebooks must have outputs stripped before commit (nbstripout enforced by CI)
- make test does NOT exist – use make test-mac on macOS
- investigation/ and kiro-*/ dirs are excluded from all CI checks and notebook operations
- ruff per-file-ignores for notebooks: E402, E501, F403, F405, E722, F821, E741, F841 (pyproject.toml)
- AgentCore containers require linux/arm64 platform and running Docker daemon
- Terraform samples require terraform init before terraform apply
- Verify factual claims against actual source files before asserting them

Action 2 — Update Steering File

Expand `~/kiro/steering/tools/aws-samples-repo.md` to include all 8 service directories and the correct tech stack. Currently it only has the repo path and git remote.

Action 3 — Sanitize `.bedrock_agentcore.yaml`

Replace hardcoded account ID and IAM role ARNs with placeholders:

```
sed -i '' 's/<REAL_ACCOUNT_ID>/<ACCOUNT_ID>/g' bedrock-agentcore/strands-  
agents/.bedrock_agentcore.yaml
```

Verify the file contains no real account IDs before any `git commit`.

Common CLI Commands

Run CI checks locally (macOS):

```
make test-mac
```

Lint only:

```
make lint
```

Validate notebooks:

```
make validate
```

Strip notebook outputs:

```
nbstripout <NOTEBOOK_PATH>
```

Deploy AgentCore (requires Docker running, linux/arm64):

```
cd bedrock-agentcore/strands-agents && agentcore deploy
```

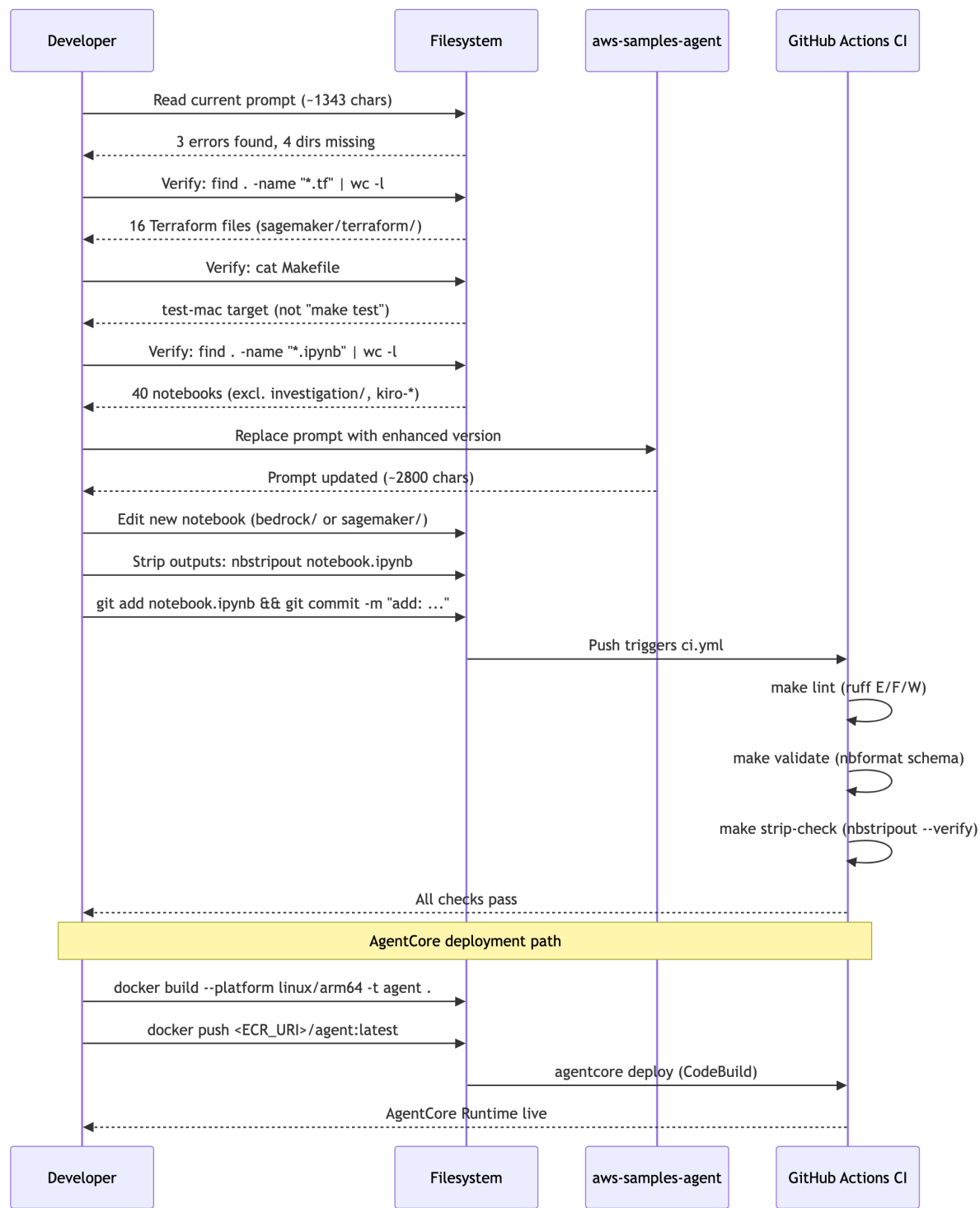
Apply Terraform IaC:

```
cd sagemaker/terraform && terraform init && terraform apply
```

Build and push SageMaker custom container:

```
docker build --platform linux/amd64 -t <IMAGE_NAME> . && docker tag  
  <IMAGE_NAME> <ECR_URI>/<IMAGE_NAME>:latest && docker push <ECR_URI>/  
  <IMAGE_NAME>:latest
```

Prompt Enhancement Sequence



Summary Table

Finding	Severity	Status	Action
Wrong tech stack in prompt (SAM/CDK)	HIGH	Confirmed	Replace prompt

Finding	Severity	Status	Action
Wrong make target (<code>make test</code>)	HIGH	Confirmed	Replace prompt
4 directories missing from prompt	HIGH	Confirmed	Replace prompt
Real AWS account ID in <code>.bedrock_agentcore.yaml</code>	HIGH	Confirmed	Sanitize file
Steering file incomplete	MEDIUM	Confirmed	Expand steering file
40 notebooks (not ~25)	LOW	Confirmed	Replace prompt

References

Source	URL
AgentCore Overview	https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/what-is-bedrock-agentcore.html
AgentCore Quotas	https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/bedrock-agentcore-limits.html
Bedrock Converse API	https://docs.aws.amazon.com/bedrock/latest/userguide/conversation-inference.html
Glue Limitations	https://docs.aws.amazon.com/glue/latest/dg/limitations.html
DataZone Quotas	https://docs.aws.amazon.com/datazone/latest/userguide/datazone-limits.html